# UK University Integration Bee Participant Guide Round 2

Vishal Gupta and Camille Mau

## Introduction

Congratulations for making it to Round 2! I hope you have fun taking part. It'll largely be the same as last year for those who took part last year.

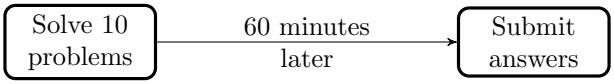
## How the competition will work

Round 2 is split into 3 rounds; a group round, a crossnumber round and a shuttle round - for those of you who did UKMT's Senior Team Maths Challenge that would be familiar. You can find examples of each round on the website too. The way each of these works is as follows:

### Group Round: 1 hour

Maximum score: 60 points.

This works the same as Round 1. Each team will be given a set of 10 problems and 60 minutes to solve as many as possible. The problems vary in difficulty but are in increasing order approximately.




Submit your answers at the end of the round. Scoring for each problem is as follows.

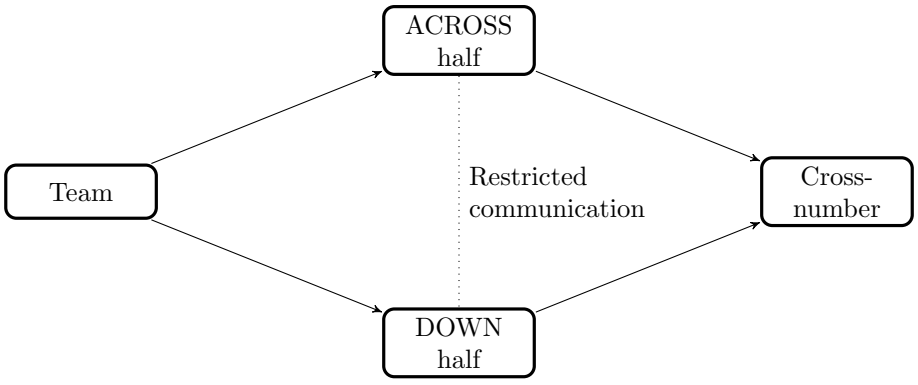
- Correct answer: 6 points
- Incorrect answer or blank: 0 points

### Crossnumber Round: 60 minutes

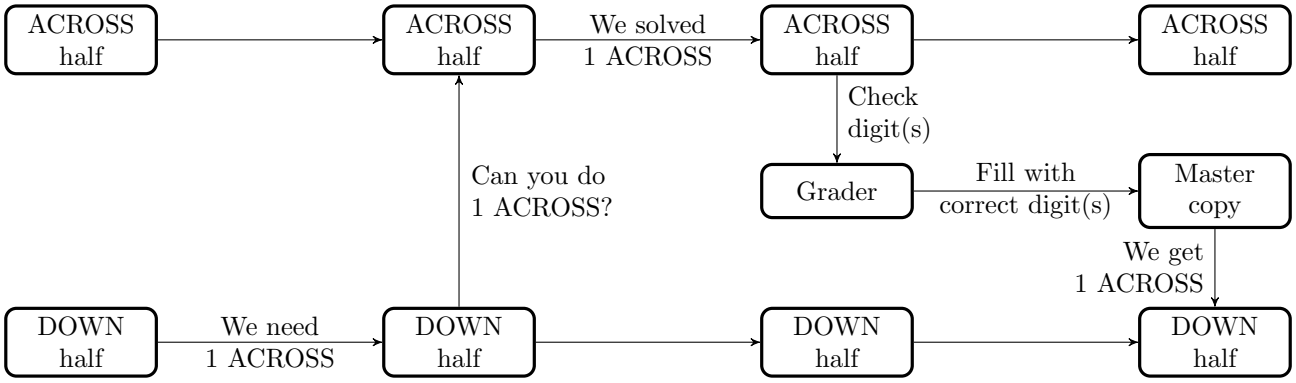
Maximum score: Varies, but typically between 45 and 50 points inclusive.

A crossnumber is essentially a crossword, but each square is a digit (0-9) instead of a letter. In this round, you are given a crossnumber puzzle and the aim is to fill as many squares as you can in 40 minutes.

The team splits into two - for a team of two, that would be a 1-1 split, and for a team of three, that would be a 2-1 split. One half gets ACROSS clues and the other half gets DOWN clues. Some of these you can do straight away, some of them you'll need to fill in numbers from other clues, and some of them depend explicitly on other clues. Clues are not necessarily all integrals. One set of clues will be intentionally made easier than the other. Teams can choose who gets each set of clues. For teams of three, we recommend giving this set to the team member on their own, but the choice is yours. A grader will also be assigned to your team and will sit with you.



No communication is allowed between the two halves of the team other than asking to do a certain clue. For example, you might be working on 8 DOWN which requires 1 ACROSS - you can ask the other half to work on 1 ACROSS. **However, the other half cannot tell you what the answer is.** Instead, the other half will have their answer graded, and the grader will then fill in the correct digit(s) in a master copy of the crossnumber grid placed between both halves of the team. Refer to this master copy for the correct answer. **Note that the only person who should write on the master copy is your grader.** The following flow illustrates a request for an answer.




Your answers will be checked as you work on the crossnumber. To submit an answer, you inform your grader that you want to submit an answer. You can choose to answer in one of the following ways.

- Square-by-square (e.g. "I want to answer this square [indicates with finger]", or "I want to answer 1 ACROSS, first square").

  In this case, you provide a digit only for the indicated square and your grader will check your answer for only that square.

- Collection of squares (e.g. "I want to answer 1 ACROSS").

  In this case, you provide the digits for that collection of squares all at once and your grader will check your answer for all squares in that collection. If this way is chosen, you cannot withdraw your answer midway through checking.

**Note that only one attempt per square is allowed.** Scoring for each square is as follows.

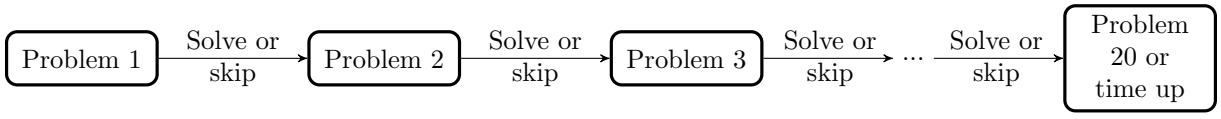
- Correct digit: 1 point
- Incorrect digit: 0 points
- Blank square at the end of the round: 0 points

Whether you get it right or wrong, once a square is attempted, the grader will write the correct digit for that square in the master copy, and that square can no longer be attempted. It's best to check one square at a time in case you mess up instead of checking the whole number at once.

## Relay Round: 45 minutes

Maximum score: 70 points.

In this round, there will be 20 problems to attempt as a team. Problems are given sequentially, one at a time. If you can't solve it you can skip it. The next problem will only be given after you solve or skip the current problem. You cannot go back to any previous problem, even if you reach the end of the 20 problems.




Each team will be assigned a grader, who will also be in charge of giving you each problem. Scoring for each problem is as follows.

- Correct answer on the first attempt: 3 points

- Correct answer on the second or later attempt: 1 point
- Skip: 0 points

A bonus to your score for this round will also be awarded based on how many problems you solve correctly (regardless of number of attempts).

- 0-9 problems: 0 points
- 10-14 problems: $\beta$ which is worth 5 points
- 15-20 problems: $\alpha$ which is worth 10 points